



REALTEK

Software Development Kit

Switch Product

Linux Tool Chain Installation Guide

(CONFIDENTIAL: Development Partners Only)

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USING THIS DOCUMENT

This document is intended for use by the system engineer when integrating with Realtek Switch Software SDK. Though every effort has been made to assure that this document is current and accurate, more information may have become available subsequent to the production of this guide. In that event, please contact your Realtek representative for additional information that may help in the development process.

REVISION HISTORY

Revision	Release Date	Summary
1.0.0	2010/6/25	Initialize the document

Table of Contents

Chapter 1	Installing Tool Chains	4
1.1.	Tool Chains for Linux Solution	4

Chapter 1 Installing Tool Chains

1.1. Tool Chains for Linux Solution

- Suggest Linux build environment
 - Currently we use CentOS Linux, which is a community-supported, free and open source operating system based on Red Hat Enterprise Linux.
- Tool chain will be a single tar-ball file.
 - rtk-ms-linux-toolchain-0.9.28.tar.gz
- Installation Steps:
 - 1. Prepare a Linux building machine, here we take CentOS Linux 5.2 as an example, gcc and all necessary library are expected to be installed.
 - ◆ For example:
[rtk-ms-linux-toolchain-0.9.28.tar.gz](#)
 - 2. Extract the tar-ball of tool chain to specified path.
 - ◆ Default:
[/local/toolchain/mipssde/mips-linux-uclibc/](#)
 - 3. Get extract source code. And open the file by an editor.
 - ◆ `kernel/linux/vendors/Realtek/SDK/config.arch`
 - 4. Make the path of tool chain which extracted on step 1 is set to following variables: CROSS_COMPILE and CROSS_KERNEL_COMPILE.
 - ◆ CROSS_COMPILE =
[/local/toolchain/mipssde/mips-linux-uclibc/bin/mips-linux-uclibc-](#)
 - ◆ CROSS_KERNEL_COMPILE =
[/local/toolchain/mipssde/mips-linux-uclibc/bin/mips-linux-uclibc-](#)

Note: Tool chain should automatically compile, if encounter any missing file of library, please install the packages required.

Note: If there is "non-authorization error", please check file: /bin/mknod, which should be suited set.

- ~~-rwsr-sr-x~~ 1 root root 26192 May 24 2008 /bin/mknod

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